DS360 - Project Phase 2

WAN 2 Configuration - Detailed Solution

1. Project Objective

The purpose of this part is to build and configure WAN 2 by creating two local area networks, LAN 3 and LAN 4, connecting the required devices, assigning correct IPv4 addresses, and verifying that the devices can communicate through Router1. The configuration follows the addressing and device requirements provided in the assignment file.

2. Required Network Topology

- Router1: Cisco 1941 router.
- Switch2 and Switch3: Cisco 2960-24TT switches.
- End devices: PC6, PC7, PC8, PC9, PC10, and PC11.
- LAN 3 is connected to one Router1 Gigabit Ethernet interface.
- LAN 4 is connected to the second Router1 Gigabit Ethernet interface.
- Recommended distribution: PC6-PC8 in LAN 3 and PC9-PC11 in LAN 4. This host distribution is a practical implementation because the source file lists six PCs but does not assign each PC to a specific LAN.

Logical layout: PC6-PC8 -> Switch2 -> Router1 <- Switch3 <- PC9-PC11

3. Physical Connections

Use Copper Straight-Through cables for both PC-to-switch and switch-to-router connections. After connecting the devices, wait until the link indicators become green before starting IP configuration.

From	Port	To	Port
PC6	FastEthernet0	Switch2	Fa0/1
PC7	FastEthernet0	Switch2	Fa0/2
PC8	FastEthernet0	Switch2	Fa0/3
Switch2	GigabitEthernet0/1	Router1	GigabitEthernet0/0
PC9	FastEthernet0	Switch3	Fa0/1
PC10	FastEthernet0	Switch3	Fa0/2
PC11	FastEthernet0	Switch3	Fa0/3
Switch3	GigabitEthernet0/1	Router1	GigabitEthernet0/1

4. IPv4 Addressing Plan

The assignment specifies the LAN 3 and LAN 4 networks and their Router1 gateway addresses. The individual PC addresses below are a proposed addressing plan that stays inside those required /24 networks and avoids duplicate addresses.

Device	LAN	IPv4 Address	Subnet Mask	Default Gateway
Router1 G0/0	LAN 3	172.16.30.1	255.255.255.0	N/A

PC6	LAN 3	172.16.30.2	255.255.255.0	172.16.30.1
PC7	LAN 3	172.16.30.3	255.255.255.0	172.16.30.1
PC8	LAN 3	172.16.30.4	255.255.255.0	172.16.30.1
Router1 G0/1	LAN 4	172.16.40.1	255.255.255.0	N/A
PC9	LAN 4	172.16.40.2	255.255.255.0	172.16.40.1
PC10	LAN 4	172.16.40.3	255.255.255.0	172.16.40.1
PC11	LAN 4	172.16.40.4	255.255.255.0	172.16.40.1

5. Step-by-Step Router1 Configuration

Open Router1, choose the CLI tab, and enter the following commands. These commands configure GigabitEthernet0/0 for LAN 3 and GigabitEthernet0/1 for LAN 4.

```
enable
configure terminal
interface gigabitEthernet0/0
 ip address 172.16.30.1 255.255.255.0
 no shutdown
exit
interface gigabitEthernet0/1
 ip address 172.16.40.1 255.255.255.0
 no shutdown
end
show ip interface brief
```

Expected result: GigabitEthernet0/0 should show IP address 172.16.30.1 and GigabitEthernet0/1 should show 172.16.40.1. Both interfaces should become up/up when the cables and connected switches are active.

6. End-Device Configuration

For each PC in Packet Tracer: click the PC -> Desktop -> IP Configuration -> select Static -> enter the IPv4 address, subnet mask, and default gateway from the addressing table above.

1. Configure PC6, PC7, and PC8 using addresses from the 172.16.30.0/24 network and gateway 172.16.30.1.
2. Configure PC9, PC10, and PC11 using addresses from the 172.16.40.0/24 network and gateway 172.16.40.1.
3. Make sure no two devices use the same IPv4 address.
4. Check that the subnet mask on all PCs is 255.255.255.0.

7. Verification and Testing

Testing is important because it proves that the topology, addressing, and Router1 interfaces are working correctly.

7.1 Verify Router Interfaces

```
show ip interface brief
```

Both LAN interfaces should appear as up/up. If an interface is administratively down, return to interface configuration mode and enter no shutdown.

7.2 Test Each PC to Its Default Gateway

On each PC, open Desktop -> Command Prompt and ping the corresponding gateway. Examples:

```
PC6> ping 172.16.30.1
PC9> ping 172.16.40.1
```

A successful reply confirms that the PC is correctly connected to its switch and that the router interface is reachable.

7.3 Test Communication Between LAN 3 and LAN 4

Because both networks are directly connected to Router1, Router1 can route traffic between them automatically. Test communication from a LAN 3 PC to a LAN 4 PC, for example:

```
PC6> ping 172.16.40.2
PC9> ping 172.16.30.2
```

Expected result: replies should be received with 0% packet loss after ARP is learned. The first ping may occasionally time out once while address resolution occurs; repeat the ping if necessary.

8. Troubleshooting Checklist

- Check that every cable is Copper Straight-Through and connected to the intended ports.
- Check that Router1 G0/0 uses 172.16.30.1/24 and G0/1 uses 172.16.40.1/24.
- Check that the no shutdown command was entered on both router interfaces.
- Check every PC IP address, subnet mask, and default gateway for typing mistakes.
- Check that LAN 3 PCs use gateway 172.16.30.1 and LAN 4 PCs use gateway 172.16.40.1.
- Check for duplicate IP addresses.
- Use show ip interface brief to confirm the router interfaces are up/up.
- If a cross-LAN ping fails, first test the PC-to-gateway ping to identify which side has the problem.

9. Evidence / Screenshots to Upload

The assignment specifically asks for evidence that matches the actual Packet Tracer work. Capture clear screenshots after the configuration is complete.

- WAN 2 topology showing Router1, Switch2, Switch3, and PC6-PC11.
- LAN 3 and LAN 4 physical connections.
- IPv4 configuration window for at least one PC in LAN 3 and one PC in LAN 4; additional screenshots can be used if required.
- Router1 CLI showing the configured LAN interfaces and/or the show ip interface brief output.
- For stronger verification, include successful ping results between LAN 3 and LAN 4 if your instructor accepts additional evidence.

10. Suggested GitHub Work Sequence

The assignment gives suggested commit descriptions. Use commit messages only when they describe changes you actually made. Do not create false or backdated project history.

Stage	Actual Work	Suggested Commit Message
Topology	Add Router1, Switch2, Switch3, and PC6-PC11; arrange LAN 3 and LAN 4.	Create WAN2 network topology
Connections	Connect PCs to switches and switches to Router1 using straight-through cables.	Add WAN2 LAN connections
Addressing	Configure Router1 LAN addresses and static IPv4 settings on the PCs.	Configure WAN2 IP addressing

11. Final Expected Result

At the end of this task, WAN 2 should contain two functioning LANs. LAN 3 should use network 172.16.30.0/24 with Router1 gateway 172.16.30.1, and LAN 4 should use network 172.16.40.0/24 with Router1 gateway 172.16.40.1. All six PCs should have valid static IPv4 settings, Router1 interfaces should be up/up, each PC should reach its default gateway, and devices in LAN 3 should be able to communicate with devices in LAN 4 through Router1.